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(c) Prove that for every continuous function f on $[0, 1]$, the sequence of polynomials $B_n f \rightarrow f$ uniformly on $[0, 1]$, where $(B_n f)$ is the sequence of Bernstein's polynomials for the function f .

[This question paper contains 8 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 4064

Unique Paper Code : 2352012401

Name of the Paper : Sequence and Series of Functions

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt any **two** parts from each question.
3. **All** questions are compulsory and carry equal marks:

1. (a) Define uniform convergence of a sequence of functions (f_n) defined on $A \subseteq \mathbb{R}$ to $\mathbb{R}$. If $A \subseteq \mathbb{R}$ and $\phi: A \rightarrow \mathbb{R}$ then define the uniform norm of

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ϕ on A . Discuss the uniform and pointwise convergence of the sequence (f_n) , where

$$f_n(x) = \frac{x}{n} \text{ for } x \in \mathbb{R} \text{ and } n \in \mathbb{N}.$$

(b) Let (f_n) be the sequence of functions defined by

$$f_n(x) = \frac{1}{1+x^n} \quad \forall x \in [0,1], n \in \mathbb{N}.$$

Find the pointwise limit of the sequence (f_n) . Does (f_n) converge uniformly? Justify your answer.

(c) Show that if (f_n) and (g_n) are two sequences of bounded functions on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ that converge uniformly to f and g respectively then prove that the product sequence $(f_n g_n)$ converge uniformly on A to fg . Give an example to show that in general the product of two uniformly convergent sequence may not be uniformly convergent.

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6. (a) For cosine function $C(x)$ and sine function $S(x)$ prove the following:

(i) If $f: \mathbb{R} \rightarrow \mathbb{R}$ is such that $f''(x) = -f(x)$ for $x \in \mathbb{R}$ then there exist real numbers α and β such that $f(x) = \alpha C(x) + \beta S(x)$ for $x \in \mathbb{R}$.

(ii) If $x \in \mathbb{R}$, $x \geq 0$ then

$$1 - \frac{1}{2}x^2 \leq C(x) \leq 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4.$$

(b) State Abel's Theorem. Show that

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \quad -1 \leq x \leq 1$$

$$\text{and } \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

(c) Show that the sequence $\left(\frac{x^n}{1+x^n}\right)$ does not converge uniformly on $[0,2]$.

3. (a) State and prove Weierstrass M-Test for uniform convergence of series of functions.

(b) Show that the series $\sum_{n=1}^{\infty} \sin\left(\frac{x}{n^2}\right)$ is uniformly convergent on $[-a, a]$, $a > 0$ but is not uniformly convergent on $\mathbb{R}$.

(c) Discuss the pointwise convergence of the series

of functions $\sum_{n=1}^{\infty} \frac{x^n}{2+3x^n}$ for $x \geq 0$.

4. (a) Let f_n be continuous function on $D \subseteq \mathbb{R}$ to $\mathbb{R}$ for each $n \in \mathbb{N}$ and $\sum_{n=1}^{\infty} f_n$ converges uniformly to f on D . Prove that f is continuous on D .

(b) Show that the series of functions $\sum_{n=1}^{\infty} \frac{\cos(x^2+1)}{n^3}$ converges uniformly on $\mathbb{R}$ to a continuous function.

(c) Given the Riemann integrable functions $f_n: \mathbb{R} \rightarrow \mathbb{R}$, $n \in \mathbb{N}$, such that

$$f_n(x) = \sin\left(\frac{x}{n^4}\right) \text{ for all } x \in \mathbb{R}$$

Show that $\sum_{n=1}^{\infty} \int_{-2\pi}^{2\pi} f_n(x) dx = \int_{-2\pi}^{2\pi} \sum_{n=1}^{\infty} f_n(x) dx$.

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5. (a) Define the radius of convergence and interval of convergence of power series. Check the uniform convergence of the following power series on $[-1, 1]$

$$x + \frac{x^2}{2^2} + \frac{x^3}{3^2} + \frac{x^4}{4^2} + \dots$$

- (b) State and Prove Cauchy-Hadamard Theorem.

- (c) Suppose that $f(x) = \sum_{n=0}^{\infty} a_n x^n$ has radius of convergence $R > 0$. Then prove that the series

$$\sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} \text{ has also radius of convergence } R$$

and for $|x| < R$

$$\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$$

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2. (a) Let (f_n) be a sequence of integrable functions on $[a, b]$ and suppose that (f_n) converges uniformly to f on $[a, b]$. Show that f is integrable on $[a, b]$ and

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n$$

- (b) Let $f_n(x) = \frac{x^n}{n}$ for $x \in [0, 1]$. Show that the sequence (f_n) of differentiable functions converges uniformly to a differentiable function f on $[0, 1]$ and that the sequence (f'_n) converges on $[0, 1]$ to a function g but the convergence is not uniform.